

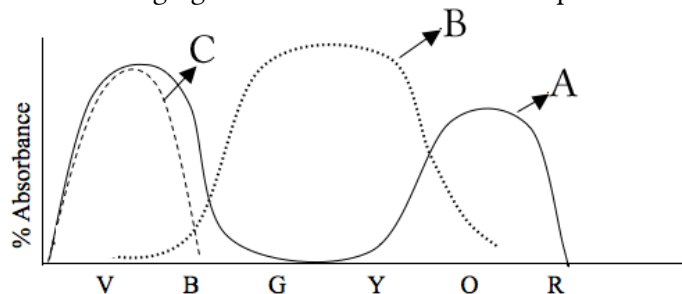
Relax! This test is not nearly as important as the start of Game of Thrones in nine days!

Each question is worth 2 points except where noted. Good luck!

Circle True or False. Read each statement completely before answering.

1. True **FALSE** Micronutrients are smaller molecules than macronutrients.
2. **TRUE** False A flaccid plant cell is most likely to occur in an isotonic environment.
3. **TRUE** False All organisms are considered open systems.
4. **TRUE** False The primary function of the large intestine is the absorption of water.
5. True **FALSE** Of the six organismal groups we are comparing, only the bacteria and the protists are not monophyletic.
6. True **FALSE** Cation exchange occurs when plants release CO₂ into the soil to increase the acidity.
7. True **FALSE** The sources of metabolic nitrogen in the diet are proteins and carbohydrates.
8. True **FALSE** The central vacuole is used by protists to eliminate excess water from the cell.
9. **TRUE** False The chemical energy stored in carbohydrates is a form of potential energy.
10. True **FALSE** A euryhaline species cannot survive in a hypotonic environment.

Use the following figure to answer the next three questions.

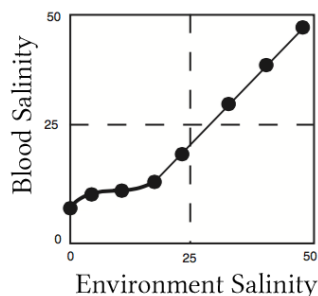


11. Line A represents the absorption spectrum of which group of photopigments? **carotenoids**
12. Line B represents the absorption spectrum of which group of photopigments? **chlorophyll**
13. Line C represents the absorption spectrum of which group of photopigments? **phycobilins**
14. (4 points) Write a *balanced* chemical equation for photosynthesis. $6 \text{CO}_2 + 6 \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 + 6 \text{H}_2\text{O}$
Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H₂O, 2 pts if more than 1 is wrong.
Subtract 1 pt if one chemical formula is wrong (e.g., C₁₂H₆O₆), 2 pts if one missing more more than 1 wrong.
Ask me if uncertain.

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Aquaporin	Hypertonic	Peristalsis
Amylase	Hypotonic	Potential
Bile	Iron bacteria	Sodium potassium pump
Chitin	Isotonic	Sphincter
Cellulose	Kinetic	Stomata
Enzymes	Methanogens	Sulfur bacteria
Flame bulb	Nephridopore	Vitamins
Glycogen	Nephridostome	Waxy cuticle
Guard cells	Pancreas	
Hydrochannel	Pepsin	

15. **Enzymes** Cofactors enhance the reaction ability of these macromolecules.
16. **Nephridostome** Opening through which body fluids enter the metanephridia.
17. **Kinetic** The energy of motion.
18. **Chitin** This structural molecule is found in cell walls and crab shells.
19. **Waxy Cuticle** Prevents water evaporation from most of the leaf surface.
20. **ANY ANSWER** Found in the digestive tract of humans.
21. **Amylase** The first digestive enzyme secreted by most animals during the digestive process.
22. **Hypertonic** The tonicity of the environment when Ψ of the environment is lower than Ψ in the organism.
23. **Aquaporin** Protein channel that increases water movement into cells.
24. The following figure shows the blood salinity level of a marine worm compared to the salinity of the surrounding environment. Based on this figure this organism is most likely an



Osmoconformer

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

25. The transfer of fluid from the glomerulus to Bowman's capsule
- A. transfers large molecules as easily as small ones.
 - B. usually includes the transfer of red blood cells into Bowman's capsule.
 - C. results from active transport.
 - D. is very selective as to which protein-sized molecules are transferred.
 - E. is mainly a consequence of blood pressure in the capillaries of the glomerulus.**
26. A marine iguana is basking on a sunny rock after feeding in the cold Pacific Ocean water. It's warming its body through
- A. convection and evaporation.
 - B. convection and radiation.
 - C. conduction and radiation.**
 - D. conduction and convection
 - E. conduction and evaporation.
27. Which of the following organs is incorrectly paired with its function?
- A. Mouth-starch digestion.
 - B. Stomach-protein digestion.
 - C. Small intestine-nutrient absorption.
 - D. Pancreas-enzyme production.
 - E. Large intestine-bile production.**
28. A species of rabbit that follows Bergmann's rule
- A. would be smaller in Minnesota than in Florida.
 - B. would be whiter in winter and browner in summer.
 - C. would be larger in Canada than in Texas.**
 - D. would have larger ears in Maine and smaller ears in Georgia.
29. Cell walls are found in at least some representatives of
- A. algae
 - B. protists
 - C. plants
 - D. bacteria
 - E. fungi
 - F. all of the above.**
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30. The osmoregulatory/excretory system of a freshwater flatworm is based on the operation of
- A. ananephredia.
 - B. metanephridia.
 - C. nephrons.
 - D. protonephridia.**
 - E. Malpighian tubules.
31. Pepsin is a digestive enzyme that
- A. digests nucleic acids in the stomach.
 - B. digests polysaccharides to monosaccharides in the small intestine.
 - C. present in bile to digest lipids in the small intestine.
 - D. begins the digestion of proteins in the stomach.**
 - E. secreted by the pancreas to digest proteins in the duodenum.
32. Which of the following is *not* a major activity of the stomach?
- A. Nutrient absorption.**
 - B. HCl secretion.
 - C. Mucus secretion.
 - D. Enzyme secretion.
 - E. Mechanical digestion.
33. Which of the following statements is correct about diffusion in or out of a cell?
- A. It is an active process in which molecules move from a region of lower concentration to one of higher concentration.
 - B. It requires aquaporins in the cell wall.
 - C. It requires channel proteins in the cell membrane.
 - D. It requires an expenditure of energy by the cell.
 - E. It is a passive process in which molecules move from a region of higher concentration to a region of lower concentration.**
34. Small mammals like mice use more energy per unit mass than large mammals mostly because most of the energy is used for
- A. growing quickly.
 - B. thermoregulation.**
 - C. the basal metabolic rate.
 - D. activities like hunting.
 - E. reproduction.
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35. When the digestion and absorption of carbohydrates results in more energy-rich molecules than are immediately required by an animal, the excess is
 - A. stored in the crop.
 - B. stored as starch.
 - C. eliminated in the feces.
 - D. stored as glycogen or fat.**
 - E. stored as cellulose or chitin.
36. A fungus is a type of
 - A. chemoautotroph
 - B. autoheterotroph
 - C. photoheterotroph
 - D. chemoheterotroph**
 - E. photoautotroph
37. After ingestion by humans, the first category of macromolecules to be chemically digested by enzymes in the mouth is
 - A. nucleic acids.
 - B. proteins.
 - C. minerals.
 - D. carbohydrates.**
 - E. cholesterol and other lipids.
38. Because the foods eaten by animals are often composed largely of macromolecules, this requires the animals to have mechanisms for _____ not found in other groups.
 - A. absorption of nutrients.
 - B. chemical digestion.**
 - C. elimination of nitrogenous waste.
 - D. reabsorption of solutes from the loop of Henle.
 - E. filtration in Bowman's capsule.
39. Stomach cells are moderately well adapted to the acidity and digesting activities in the stomach by having
 - A. a sufficient colony of *H. pylori* to breakdown the cellulose from vegetables.
 - B. a high level of bile.
 - C. a thick, mucous secretion and active replacement of epithelial cells.**
 - D. a high level of secretion of pepsin.
 - E. secretions enter the stomach from the pancreas.

40. Birds secrete uric acid as their nitrogenous waste because uric acid
- A. is readily soluble in water.
 - B. requires little water for nitrogenous waste disposal, thus reducing body mass.**
 - C. is less toxic than urea.**
 - D. is metabolically less expensive to synthesize than other excretory products.
 - E. excretion allows birds to live in desert environments.
41. Which of the following is an accurate statement about thermoregulation?
- A. Endotherms maintain a constant body temperature and ectotherms do not.
 - B. Endotherms and ectotherms differ in their primary source of heat for thermoregulation.**
 - C. Endotherms are warm-blooded and ectotherms are cold-blooded.
 - D. Endothermy has a lower energy cost than ectothermy.
 - E. Endotherms are regulators and ectotherms are conformers.
42. Which of the following types of molecules are the major structural components of the cell membrane?
- A. proteins and cellulose.
 - B. phospholipids and cellulose.
 - C. nucleic acids and proteins.
 - D. phospholipids and proteins.**
 - E. proteins and collagen.
43. Upon activation by stomach acidity, the secretions of the gastric glands
- A. initiate the chemical digestion of lipids in the stomach.
 - B. include nitrogenase and amylase.
 - C. delay digestion until the food arrives in the small intestine.
 - D. initiate the digestion of protein in the stomach.**
 - E. initiate the mechanical digestion of lipids in the stomach.
44. Ingested dietary substances must cross cell membranes to be used by the body, a process known as
- A. elimination.
 - B. ingestion.
 - C. absorption.**
 - D. hydrolysis.
 - E. digestion.
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45. Which of the following membrane activities requires energy from ATP?
- A. movement of glucose molecules into a bacterial cell from a solution containing a higher concentration of glucose than inside the cell.
 - B. diffusion of chloride ions across the membrane through a chloride channel.
 - C. movement of water out of a paramecium.
 - D. movement of Na^+ ions from a lower concentration in a mammalian cell to a higher concentration in the extracellular fluid.**
 - E. movement of water into a cell.
46. Which of the following contains digestive enzymes?
- A. food vacuole.
 - B. central vacuole.
 - C. contractile vacuole.
 - D. lysosome**
 - E. guard cells.
47. The advantage of excreting nitrogenous wastes as urea rather than as ammonia is that
- A. less nitrogen is removed from the body.
 - B. urea can be exchanged for Na^+ .
 - C. urea is less toxic than ammonia.**
 - D. urea does not affect the osmoregulation.
 - E. urea requires more water for excretion than ammonia.
48. Stomata will open when
- A. the pressure potential inside the guard cell is negative.
 - B. the water potential inside the guard cell is higher than outside the cell.
 - C. the solute potential inside the guard cell is positive.
 - D. the guard cells have a higher solute concentration than outside the cell.**
 - E. the guard cells have a lower solute concentration than outside the cell.
49. Organisms that obtain energy by oxidizing iron compounds and obtain carbon from CO_2 are called
- A. chemoheterotrophs.
 - B. photochemotrophs.
 - C. photoheterotrophs.
 - D. photoautotrophs.
 - E. chemoautotrophs.**
 - F. autoheterotrophs.
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